anyone-Hub: Platform Security and Architecture Whitepaper

1. The JediTerm Native PTY Architecture

At the core of the anyone-Hub terminal execution model is our vendored integration of JetBrains' JediTerm. Rather than relying on a bloated, resource-heavy Debian VM—which introduces a massive attack surface and significant virtualization overhead—we engineered a native PTY (Pseudo-Terminal) bridge. By interfacing directly with the native Android execution environment, this architecture enables direct, on-the-metal dependency resolution and local project compilation. It drastically reduces the attack surface by eliminating a secondary OS environment. There are no exposed hypervisors, no secondary network stacks, and no latent background daemons that could be compromised.

2. The Mechanics of Zero-Copy IPC (FlatBuffers)

High-throughput Inter-Process Communication (IPC) is achieved exclusively through FlatBuffers. Our .fbs schemas define rigid data contracts between the Kotlin orchestration layers and the C++ execution engine. Crucially, FlatBuffers permits Kotlin and C++ environments to read direct byte offsets within the payload buffer without any intermediate deserialization. This zero-copy transit completely eliminates heap allocations and object parsing overhead. By operating directly on the memory map, we fundamentally neutralize memory corruption vulnerabilities, buffer overflows, and malicious deserialization attack vectors, all while achieving near-zero latency.

3. Extensions Marketplace & Ecosystem Security

The anyone-Hub plugin architecture is designed with security at its foundation. We ensure ecosystem integrity through a zero-trust model for all third-party extensions. Our Supabase integration strictly verifies the cryptographic integrity and provenance of all extensions available in the marketplace, preventing the distribution of tampered or unauthorized packages. When a third-party extension is executed, it runs within a heavily restricted, isolated classloader environment. This sandboxing mechanism explicitly strips the extension of unprivileged network access and prevents arbitrary interactions with the local host filesystem. Extensions are strictly confined to their declared operational capabilities, neutralizing the risk of supply-chain attacks.

4. Threat Mitigation Strategy

Our holistic architecture provides profound neutralization against hostile project code. A primary threat vector in IDE environments involves cloned repositories executing arbitrary, destructive commands on the host machine. To mitigate this, anyone-Hub utilizes unprivileged, contained execution boundaries via ProcessBuilder combined with strict Minijail configurations and execution whitelists. If a cloned repository contains malicious scripts intended to compromise the host Android device, the sandboxed environment prevents access to sensitive host directories, device root permissions, and core Android IPC mechanisms. The malicious code remains isolated, and its execution blast radius is contained exclusively within the unprivileged project sandbox.